

Marcus Hernandez

Marcusrhernandez99@gmail.com | (858) 209-7317 | San Diego, CA | [LinkedIn](#)

EDUCATION

San Diego Mesa College

Expected Graduation: Jun. 2027

Computer Engineering

Honors: San Diego Promise Program Scholar, Phi Theta Kappa Honors Society | **GPA: 4.0**

PROJECTS

Autonomous Go Kart (Autonomous Kart Series 2026)

Jan. 2026 - Present

Mesa Full Self-Driving Club, Lead Electrical Engineer

- Leading the electrical system design for a competition-ready autonomous go-kart, wiring servo motors, motor controllers, and sensors to STM32 and Arduino microcontrollers for steering and throttle control.
- Configured ExpressLRS receiver communication over UART to translate RC input into PWM servo commands, debugging signal timing and failsafe logic to ensure reliable autonomous and manual operation.
- Preparing the full electrical and software stack for competition in May 2026, including power distribution, sensor integration, and control system tuning for real-time autonomous navigation.

Autonomous Underwater Vehicle (RoboSub 2026)

Aug. 2025 - Present

Miramar Engineering Club

- Building an autonomous underwater vehicle for RoboSub competition using a Jetson Nano running Python 3 and ROS to process camera data and send navigation commands through a Pixhawk thruster system.
- Wrote Python diagnostic scripts and used multimeter measurements to isolate thruster and power delivery failures, developing a methodical debugging workflow for hardware–software interaction.
- Integrated cameras and power distribution into the ROS-Pixhawk stack and validated sensor feedback and vehicle response during pool-based system testing.

LEADERSHIP & CAMPUS INVOLVEMENT

Engineering Club, Miramar College

Aug. 2025 - Present

ASG Representative

- Represent the Engineering Club in student government, acting as a liaison between the club and ASG.
- Advocate for funding and institutional support for engineering projects in a resource-limited college setting.

STUDENT GOVERNMENT

Associated Student Government, Miramar College

Jan. 2026 - Present

ASG Senator (1 of 20)

- Vote on bills and constitutional amendments while holding weekly senator hours (3+ hrs/week) in the ASG Office to assist students with transfer guidance, class registration, and general questions.
- Developing a senator project focused on expanding student access to STEM resources, including research opportunities, faculty mentorship programs, and internship pipelines.
- Advocate for student needs by collaborating with fellow senators and campus administration on policy decisions and resource allocation.

SKILLS & RELEVANT COURSEWORK

Skills: Python | Java | OpenCV | ROS | Linux | Git | Arduino | STM32 | Raspberry Pi | UART/PWM | Soldering

Coursework: Calculus III | Discrete Mathematics | Intermediate Java | Physics II: Electricity & Magnetism